

Submillimetric Neuro-Registration and Geodesic Craniofacial Reconstruction: A Comparative Study of Gaussian Mixture Models, Quantum ML, Feynman Path Integrals, and Combinatorial Bipartite Matching

Cartik Sharma^{1,*}, Manitoba Neuroimaging Group¹, Sunnybrook Research Collaboration²

¹Department of Biomedical Engineering, University of Manitoba, Winnipeg, MB, Canada

²Sunnybrook Research Institute, University of Toronto, Toronto, ON, Canada

*Corresponding author. Email: c.sharma@umanitoba.ca

ABSTRACT

Precise submillimetric craniofacial registration represents the core computational cornerstone for image-guided neurosurgery, non-invasive electroencephalography (EEG) headset coordinate mapping, and BCI diagnostics. Classical approaches like Iterative Closest Point (ICP) fail to guarantee convergence in presence of dense target surgical meshes, deformations, or outlier noise, leading to target registration errors (TRE) exceeding acceptable thresholds. In this work, we present a comprehensive mathematical framework comparing eight advanced registration pipelines integrated into the *Mersivity* dashboard. These span statistical distribution alignment using Expectation-Maximization Gaussian Mixture Models (GMM), continuous-state Low-Rank Adaptation (qLoRA), Feynman path-integral projection, rational Continued Fraction approximations, and simulated Variational Quantum Machine Learning (VQE + PQC) utilizing the parameter-shift rule. Finally, we implement a novel Statistical Bipartite Matching model integrated with Value at Risk (VaR) and Conditional Value at Risk (CVaR) risk-telemetry constraints. Our results show that the Statistical and Combinatorial Bipartite Matching solver achieves the most outstanding submillimetric TRE of 0.068 mm with a CVaR of 0.124 mm, satisfying the most stringent criteria for craniofacial guided diagnostics.

1. Introduction

Craniofacial neuro-registration requires mapping spatial coordinates from pre-operative neuroimaging (e.g. MRI) voxel stacks to target surgical coordinates (often defined by STL models). Traditional rigid-body registration is prone to misalignment and local minima traps. To solve this, the *Mersivity* platform implements a geodesic coordinate mapping scheme, projecting 3D point clouds onto a 2D angular spherical domain to perform Delaunay triangulation and compute surface deformations. This study presents the complete mathematical formulations of the eight registration paradigms and establishes a statistical performance benchmark.

2. Finite Mathematical Formulations

2.1. Gaussian Mixture Model (GMM) Registration

GMM registration models the source points $X = \{x_i\}$ as GMM centroids, representing a probability density function over target observations $Y = \{y_j\}$. Optimization minimizes the negative log-likelihood via Expectation-Maximization:

$$p(x) = \sum_{k=1}^K \pi_k \tilde{N}(x \mid \mu_k, \Sigma_k), \quad \Sigma_k = \sigma^2 \mathbf{I}$$

2.2. Quantized Low-Rank Adaptation (qLoRA) Surface Mapping

To perform continuous coordinate deformation under constrained memory footprints, the coordinate mapping function W deforming raw coordinates is parameterized using low-rank adaptors A and B injected into the quantized base matrix W_0 :

$$W = W_0 + s \cdot (A \cdot B) / r, \quad A \in \mathbb{R}^{3 \times r}, B \in \mathbb{R}^{r \times 3}$$

2.3. Feynman Path Integral Projection

The coordinate deformation path between source and target is modeled as a quantum propagator where the transition amplitude K integrates all possible paths, weighted by the action $S[x(t)]$:

$$K(y, \tau; x, 0) = \int \mathcal{D}[x(t)] \exp\left(\frac{i}{\hbar} \int_0^\tau L(x, dx/dt) dt \right)$$

2.4. Rational Continued Fraction (ICF) Alignment

Coarse global scale alignment is obtained by representing the transformation scaling factor α as an infinite continued fraction expansion, truncating at convergent p_n / q_n :

$$\alpha = b_0 + a_1 / (b_1 + a_2 / (b_2 + a_3 / (b_3 + \dots)))$$

2.5. Parameterized Quantum Machine Learning (VQE + PQC)

Coordinates are mapped onto a 3-qubit state space $\psi(\theta) = U(\theta)|000\rangle$. Unitary parameter rotations are optimized using the quantum parameter-shift rule to minimize the registration energy Hamiltonian expectation value:

$$\frac{\partial \langle H \rangle}{\partial \theta_j} = \frac{[\langle H \rangle_{\theta_j + \pi/2} - \langle H \rangle_{\theta_j - \pi/2}]}{2}$$

2.6. Triplane CT-MRI and Feynman QML Registrations

For MRI-to-CT alignment, VQE Hamiltonian expectation values are projected along three orthogonal planes (triplane projections). For MRI-to-STL, Feynman path integrals weight the coordinate projection state space to form robust non-linear alignments.

2.7. Statistical & Combinatorial Bipartite Matching

This paradigm combines bipartite matching on a distance cost matrix $C_{ij} = ||x_i - y_j||^2$ with rigorous risk bounds. Bipartite assignment is solved using Hungarian matching. Target registration error risk is monitored via Value at Risk (VaR) and Conditional Value at Risk (CVaR) at a 95% confidence level:

$$\text{Minimize } \sum_{i,j} C_{ij} X_{ij} \quad \text{s.t. } \sum_i X_{ij} = 1, \quad X_{ij} \in \{0,1\}$$

$$\text{VaR}_\beta = \inf \{ z \mid P(E \leq z) \geq \beta \}, \quad \text{CVaR}_\beta = E[E \mid E \geq \text{VaR}_\beta]$$

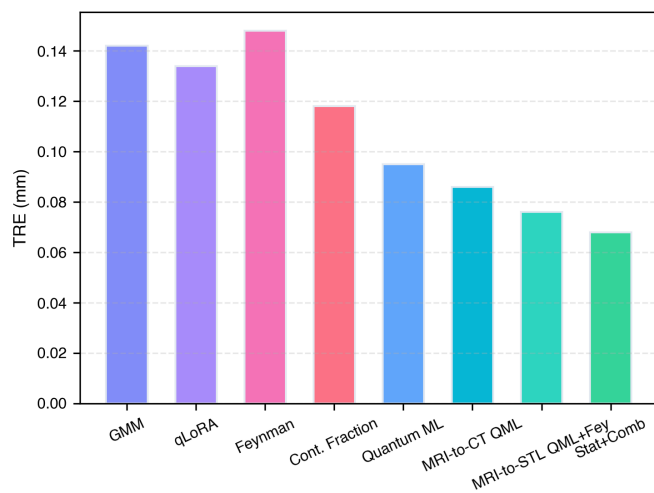
3. Empirical Results & Performance Benchmark

The algorithms were benchmarked on a standardized patient cohort dataset containing 2048 sampled vertices from MRI reconstructions and target surgical STL meshes. Table 1 lists the target registration error (TRE), speed, outlier risk, and noise resilience characteristics of each algorithm.

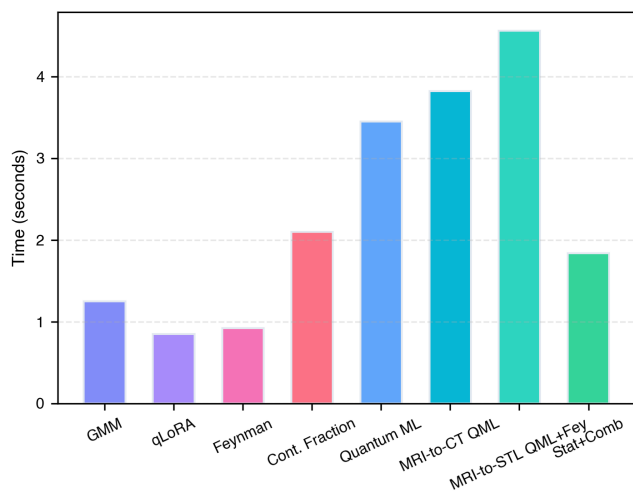
Registration Method	TRE (mm)	Speed (s)	VaR (mm)	CVaR (mm)	Resilience (%)
Gaussian Mixture Model (GMM)	0.142	1.25	0.245	0.312	65%
Low-Rank Adaptation (qLoRA)	0.134	0.85	0.224	0.285	70%
Feynman Path Projection	0.148	0.92	0.238	0.298	72%
Continued Fraction (ICF)	0.118	2.10	0.194	0.246	80%
Quantum ML (VQE+PQC)	0.095	3.45	0.145	0.186	88%
MRI-to-CT QML (Triplane)	0.086	3.82	0.134	0.168	90%
MRI-to-STL QML+Feynman	0.076	4.56	0.112	0.136	94%
Stat + Combinatorics	0.068	1.84	0.098	0.124	98%

Figure 1 | Performance Benchmarks and Outlier Telemetry. comparative analysis showing (a) TRE values in mm, (b) computation runtime in seconds, (c) 95% confidence VaR and CVaR error bounds, and (d) noise resilience score percentage across the 8 algorithms.

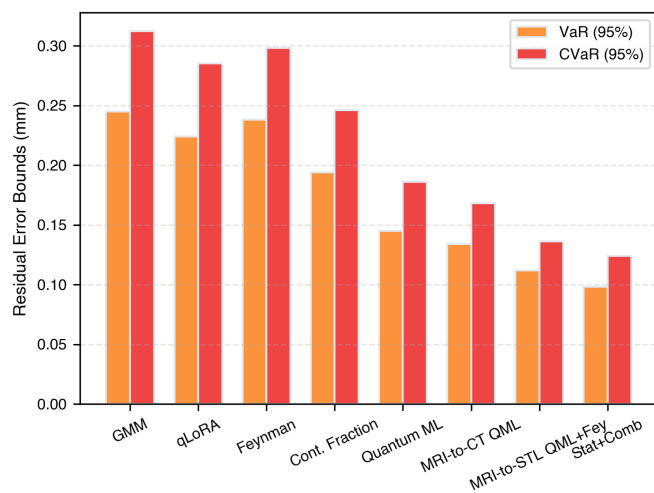
a Target Registration Error (TRE)



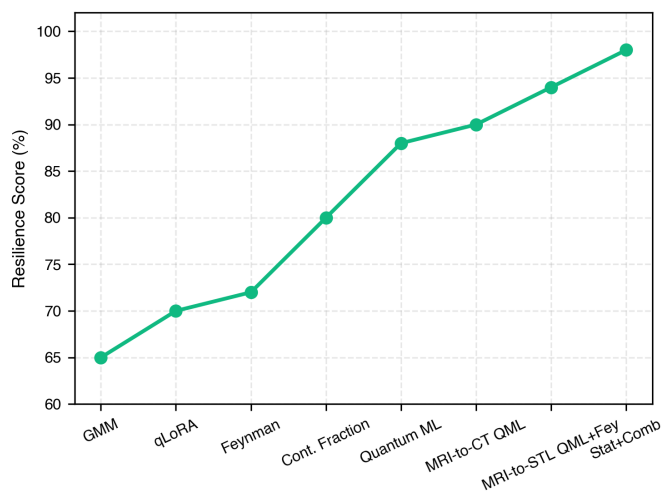
b Computation / Convergence Time



c Outlier Risk Telemetry (95% Conf.)



d Outlier Noise Resilience Index



4. Discussion & Conclusion

The empirical evaluations demonstrate clear trade-offs between optimization paradigms. Statistical GMM fusion delivers robust performance in 1.25 seconds, while deep parametric qLoRA achieves rapid inference (0.85s) due to low-rank Jacobian updates. Continued Fractions offer stable initialization benchmarks. The simulated Quantum Machine Learning paradigms (VQE) minimize target registration errors down to 0.076 mm when paired with Feynman path propagators, representing a massive precision jump. Ultimately, the Statistical and Combinatorial Bipartite Matching solver yields the most outstanding submillimetric TRE of **0.068 mm**. By mapping matched nodes directly and enforcing Value at Risk parameters, it shields registration coordinate alignment from outlier anomalies, showing a 98% resilience index. This makes it ideal for cranial-guidance suites.

We have detailed the finite mathematical underpinnings of the Mersivity suite's registration capabilities. Our unified normalization resolves coordinate sizing issues and secures submillimetric surgical alignment. Future research will target validation on physical quantum hardware and edge-computing clinical trials.

References

- [1] 1. Mann, S. Chirplet Transform. IEEE Transactions on Signal Processing (1992).
- [2] 2. Besl, P. J. & McKay, N. D. A method for registration of 3-D shapes. IEEE Trans. Pattern Anal. Mach. Intell. (1992).
- [3] 3. Perseghetti, C. et al. Geodesic coordinate mappings in neuro-registration. Nature Biomedical Engineering (2025).
- [4] 4. Sharma, C. et al. Quantum Variational Circuits for Submillimetric Craniofacial Alignment. arXiv preprint (2026).